

ANGELA WANJIRU MAINA

AI Engineer | Data Science & Machine Learning Specialist

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with expertise in Python programming, machine learning model development, and data science workflows. Strong foundation in computer science fundamentals including data structures, algorithms, and object-oriented programming (C++). Certified Data Scientist with hands-on experience building sentiment analysis systems, recommendation engines, and classification models. Proven ability to translate complex technical requirements into practical AI-driven solutions while collaborating with cross-functional teams.

TECHNICAL SKILLS

Programming & AI/ML:

- Python (NumPy, Pandas, scikit-learn, TensorFlow, Keras), R Programming
- C++ (Data Structures & Algorithms coursework) - strong foundation for learning C#/.NET ecosystem
- Machine Learning: Supervised/Unsupervised Learning, Random Forests, Logistic Regression, Naive Bayes, Decision Trees
- NLP & Text Analytics: NLTK, TF-IDF vectorization, Sentiment Analysis, Text preprocessing and feature engineering
- Model Evaluation: Cross-validation, Hyperparameter tuning (GridSearchCV), SMOTE for class imbalance, Model optimization

Data & Database Management:

- SQL (complex queries, joins, aggregations), NoSQL (MongoDB), Database design and optimization
- Data pipeline development, ETL processes, Data cleaning and preprocessing

APIs & Integration:

- RESTful API development and integration, OAuth, HTTP protocols, JSON/CSV data handling
- Experience consuming third-party APIs and building data ingestion pipelines

Development Tools & Fundamentals:

- Git/GitHub (version control, collaboration), Jupyter Notebooks, Excel VBA (automation)
- Object-Oriented Programming, Data Structures (arrays, linked lists, trees, graphs, hash tables)
- Algorithms (sorting, searching, dynamic programming), Time/Space complexity analysis

Currently Building Proficiency:

- C# and .NET Core (leveraging C++ OOP background for rapid learning)
- Microsoft Azure (ML services, App Services, Function Apps), CI/CD pipelines with GitHub Actions

PROFESSIONAL EXPERIENCE

Environment Information Officer – Air Quality & Climate Change

Nairobi City County | September 2023 – Present

- Developed automated data pipelines using Python and R, reducing manual processing time by 40% through custom scripts and Excel VBA macros
- Built and maintained air quality database management system, ensuring data accuracy and consistency across sensor networks
- Applied machine learning techniques for PM2.5 and NO2 trend analysis, utilizing statistical modeling and quantitative analysis
- Integrated data from multiple sources (AirQo sensors, reference monitors) using API connections and data transformation workflows
- Collaborated with C40 Cities, WRI, and University of Nairobi on 2025 GHG Inventory, demonstrating multi-stakeholder project coordination

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- Created data visualization dashboards and generated quarterly reports for policymakers, translating complex datasets into actionable insights

Air Quality Run Leader – Citizen Science & Advocacy

UrbanBetter Cityzens4CleanAir Campaign | April 2025 – August 2025

- Designed and executed mobile sensor data collection system, processing hyperlocal air quality data across neighborhoods
- Developed data processing pipeline to transform raw sensor data into policy-relevant reports
- Engaged 200+ residents through data-driven advocacy, demonstrating stakeholder communication and project management skills

KEY AI/ML PROJECTS & DEPLOYMENTS

Sentiment Flow – Twitter Sentiment Analysis System (Phase 5, 2024)

- Built end-to-end NLP sentiment classification system analyzing 9,000+ tweets about Apple and Google products, achieving 84% accuracy with Random Forest + TF-IDF vectorization
- Implemented comprehensive text preprocessing pipeline: tokenization, stopword removal, lemmatization, handling class imbalance with SMOTE
- Compared 4 ML models (Random Forest, Naive Bayes, Logistic Regression, Decision Trees) using both Count and TF-IDF vectorization
- Deployed production model with interactive web application for real-time sentiment prediction, providing actionable insights for product teams

Movie Recommendation Engine (Phase 4, 2024)

- Developed collaborative filtering recommendation system using matrix factorization and Alternating Least Squares (ALS) algorithm
- Implemented Singular Value Decomposition (SVD) for dimensionality reduction on user-movie rating matrices
- Built recommender system using Surprise library, optimized hyperparameters, and evaluated using RMSE and MAE metrics

Customer Churn Prediction System (Phase 3, 2024)

- Built binary classification model to predict customer churn with 95%+ accuracy using ensemble methods (Random Forest, Gradient Boosting)
- Performed feature engineering on telecommunications data, handling categorical variables through one-hot encoding and feature scaling
- Applied hyperparameter tuning using GridSearchCV and cross-validation to optimize model performance
- Generated business insights identifying key churn drivers, enabling proactive customer retention strategies

C40 Cities Air Quality Analysis (2024-2025)

- Analyzed PM2.5 and NO2 pollution trends across 96 global cities using Python, implementing statistical comparisons and time series analysis
- Built data visualization suite with health impact analysis, converting R scripts to Python for improved scalability and maintainability

EDUCATION & CERTIFICATIONS

Data Science and Machine Learning Certificate

Moringa School, Nairobi, Kenya | 2024

Air Quality Monitoring Certification

University of Nairobi, Kenya | 2024

GIS for Environment and Water Resources Certificate

IDEAMAPS, Nairobi, Kenya | 2024

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Bachelor of Science (B.Sc.) - Engineering Stream

Saint Mary's University, Halifax, NS, Canada | 2021

Diploma in Engineering

Saint Mary's University, Halifax, NS, Canada | 2021

Coursework: Data Structures & Algorithms (C++), Engineering Mathematics, Quantitative Methods

ADDITIONAL STRENGTHS

- Strong computer science fundamentals with formal training in data structures, algorithms, and computational complexity
- Experience with object-oriented programming principles (C++) providing foundation for rapid C# adoption
- Strong analytical thinking and problem-solving abilities with experience translating business requirements into technical solutions
- Excellent communication skills with experience presenting technical findings to non-technical stakeholders and government officials
- Proven ability to learn new technologies quickly - demonstrated by transitioning from C++ to Python/R to deploying ML models
- GitHub portfolio demonstrating end-to-end ML project implementation and deployment capabilities